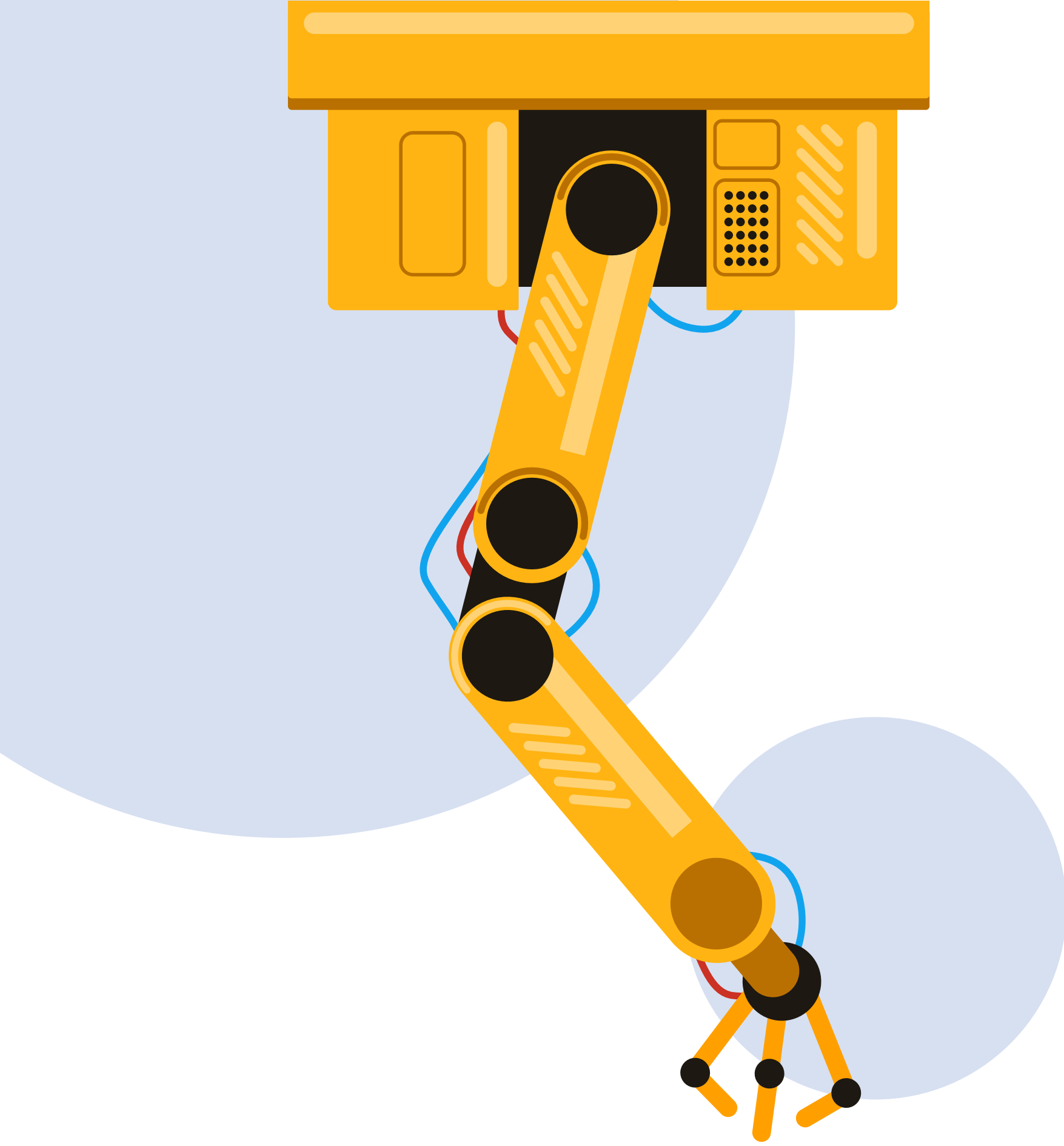


CLASE 8

Robótica





Visión artificial (Teoría)

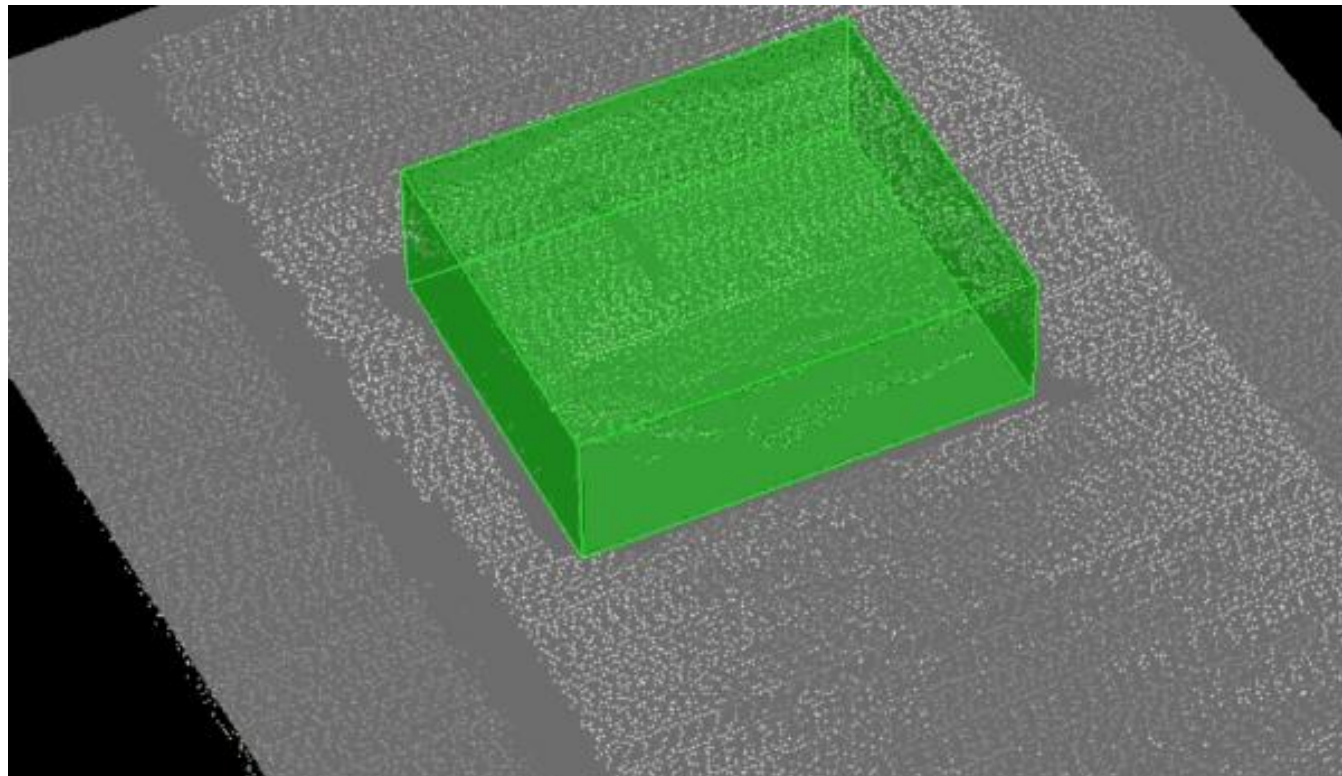
- A.1 Tipos de sistemas de visión
- A.2 Algoritmos
- A.3 Componentes
- A.4 Sistemas de iluminación

Tipos de sistemas de visión

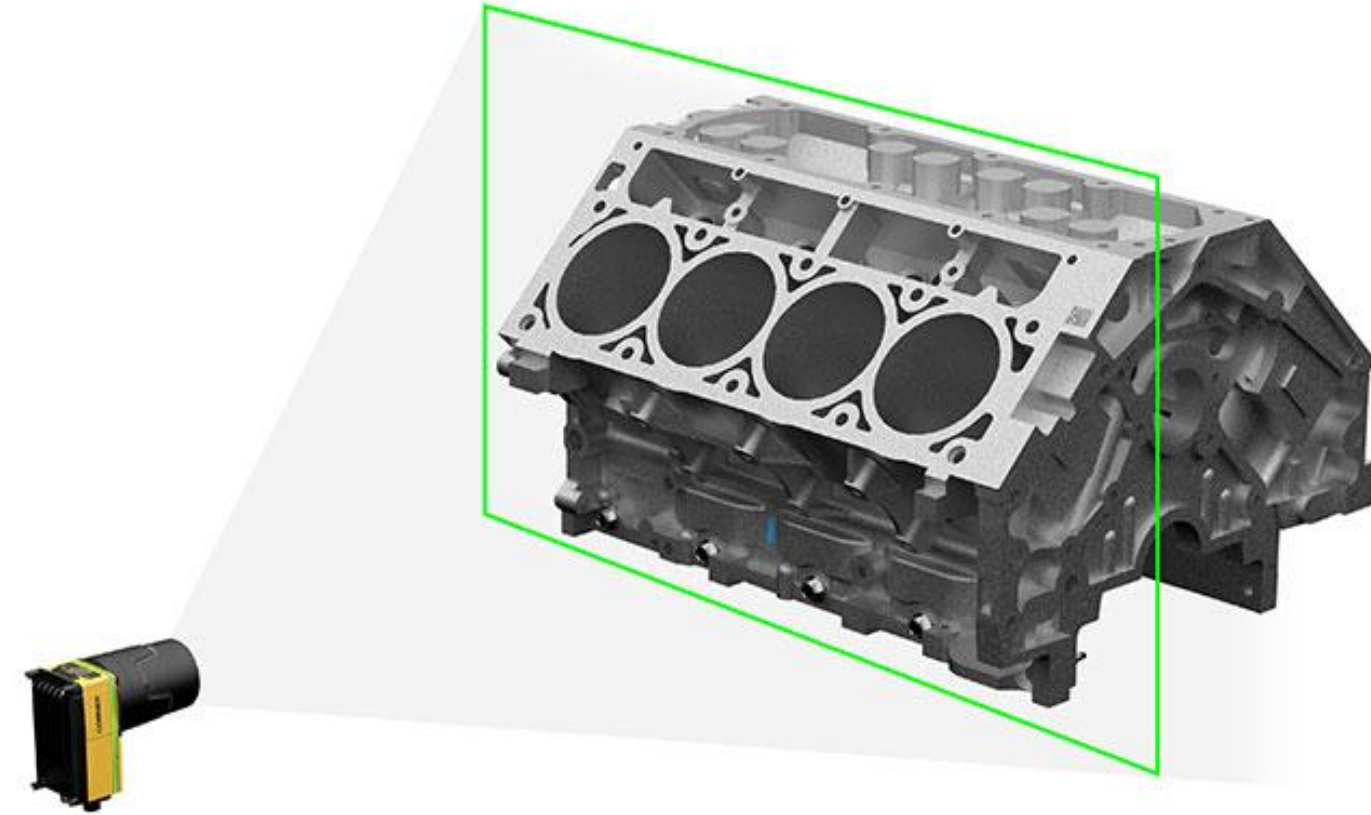
Cámaras de línea



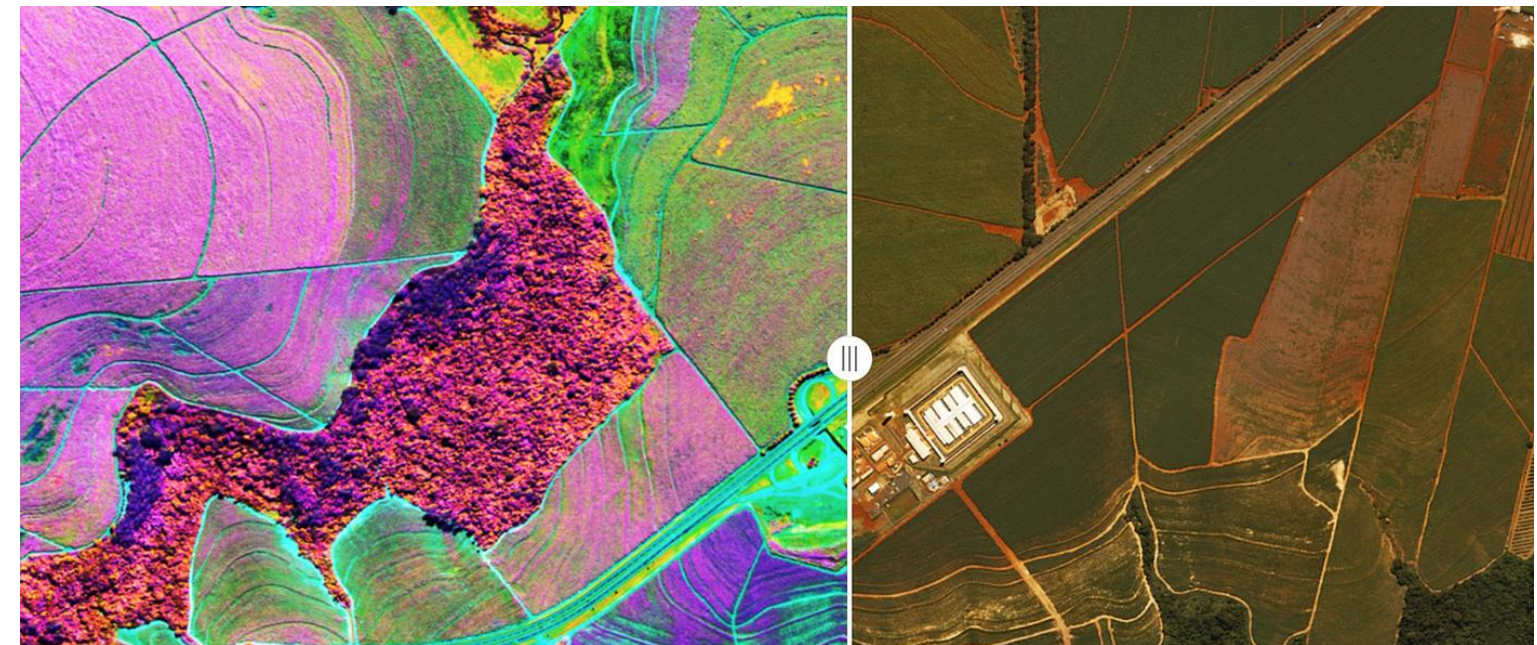
Cámaras 3D



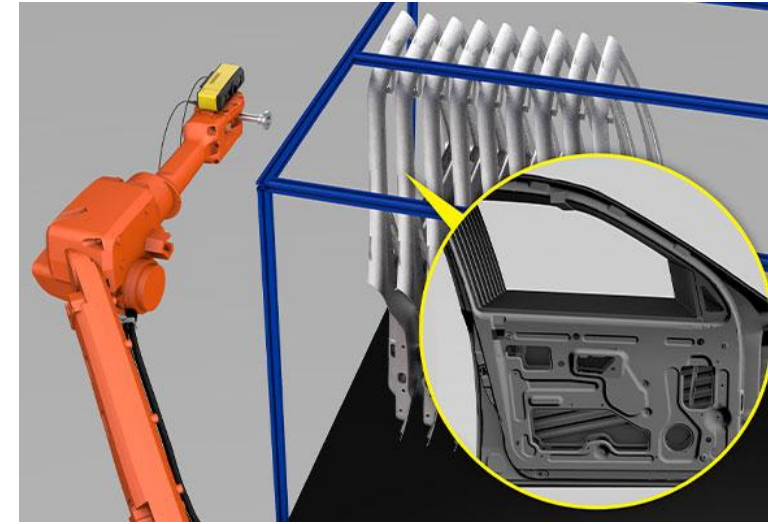
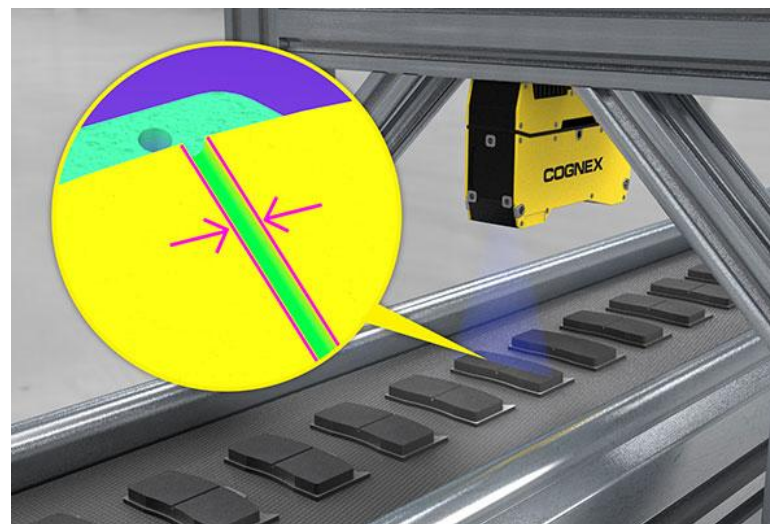
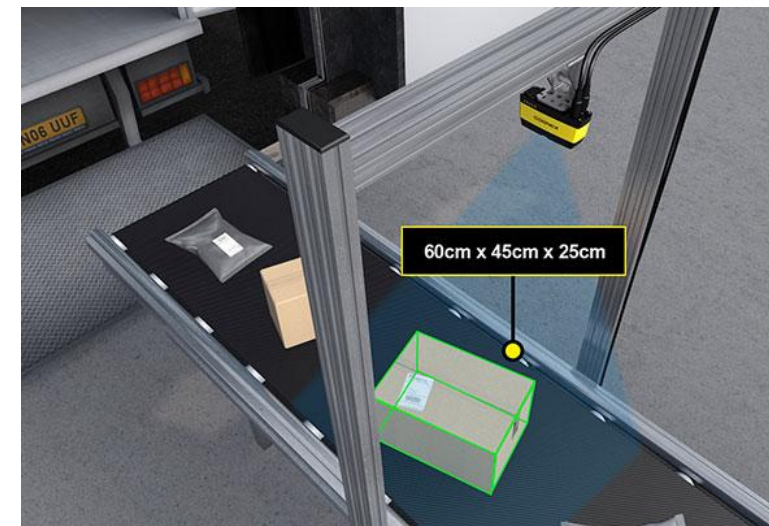
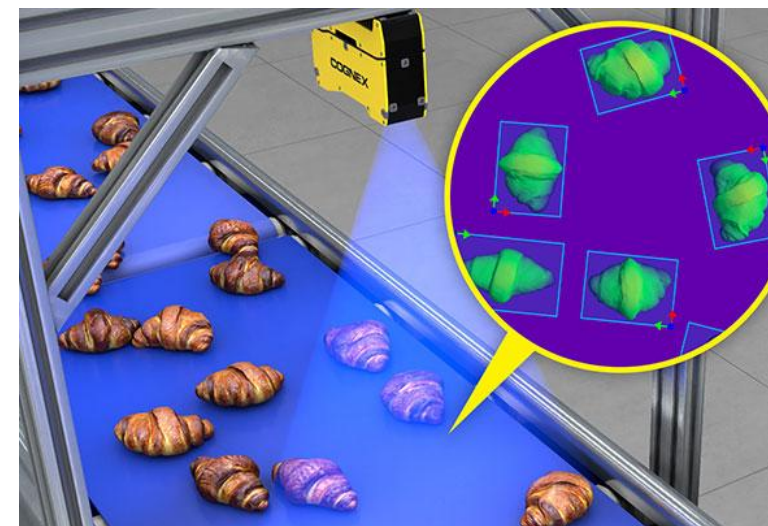
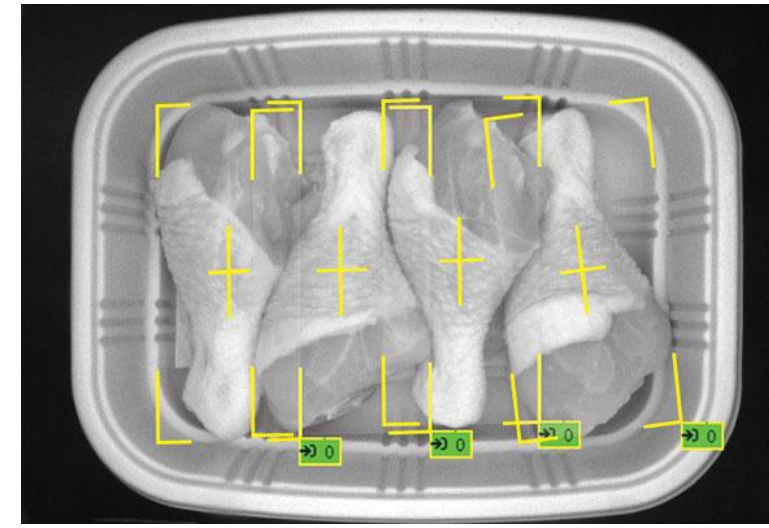
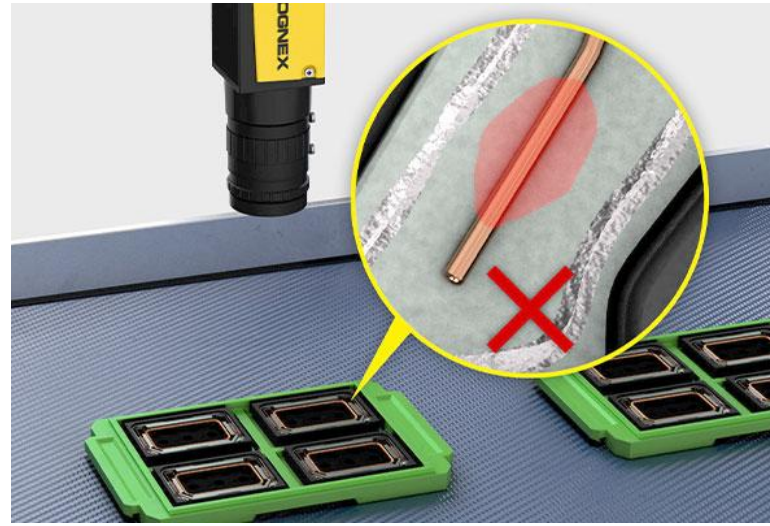
Cámaras de area



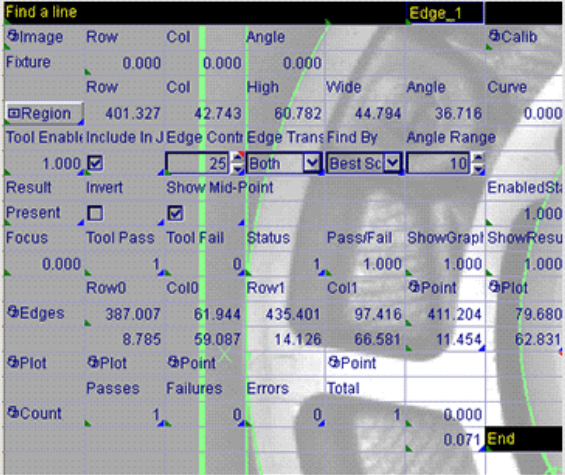
Cámaras hiperspectrales



Algoritmos



Algoritmos

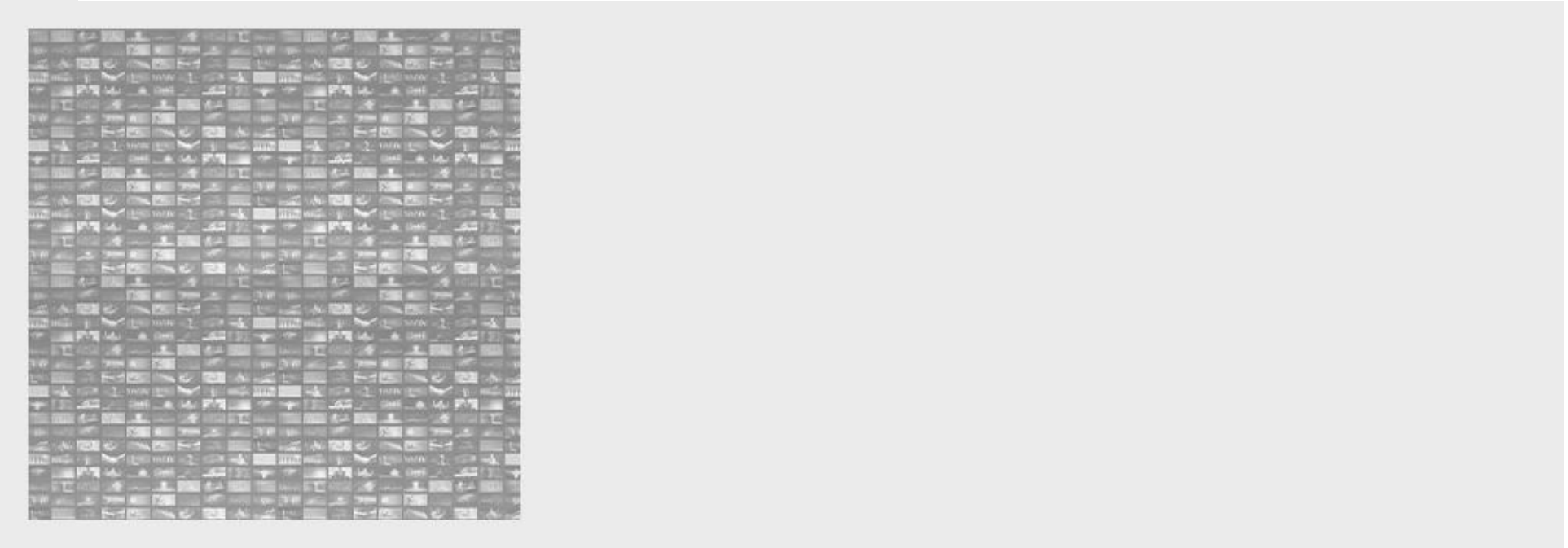


Find a line

Edge_1

Image	Row	Col	Angle	Calib
Fixture	0.000	0.000	0.000	
Row	Col	High	Wide	Angle
Region	401.327	42.743	60.782	44.794
Tool Enable	Include	In J	Edge Contr	Edge Trans
Find By	Angle Range			
1.000	☒	25	Both	Best Sc
Result	Invert	Show Mid-Point		EnabledSt
Present	☐	☒		1.000
Focus	Tool Pass	Tool Fail	Status	Pass/Fail
0.000	1	0	1	1.000
Row0	Col0	Row1	Col1	Point
Edges	387.007	61.944	435.401	97.416
	8.785	59.087	14.126	66.581
Plot	Plot	Point	Point	Point
Count	Passes	Failures	Errors	Total
	1	0	0	1
				0.000
				0.071
				End





Algoritmos



Rule-based vision



Measurement and gauging



Location and fixturing



Robotic guidance



Edge learning



Classification



Assembly verification



Character reading



Deep learning



Complex defect detection

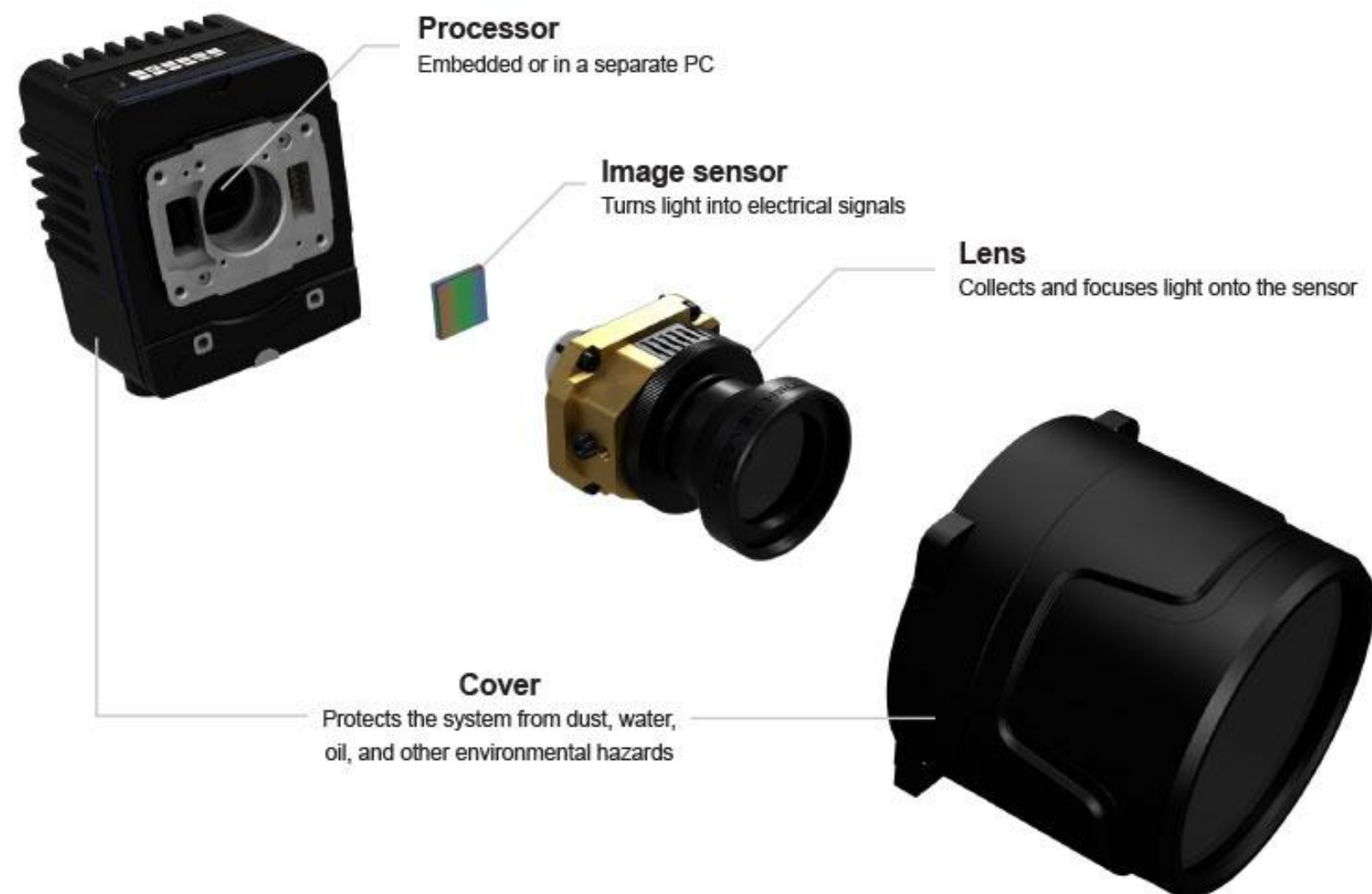


Tasks with significant variation

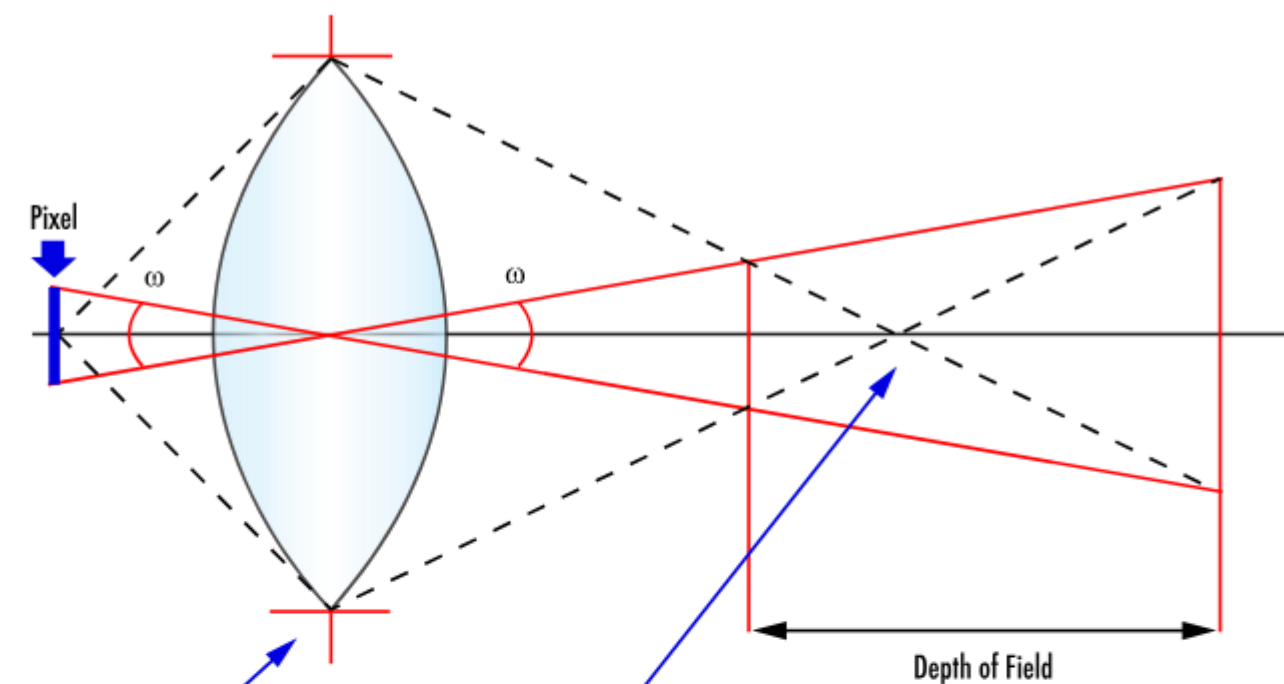


Highly customized jobs

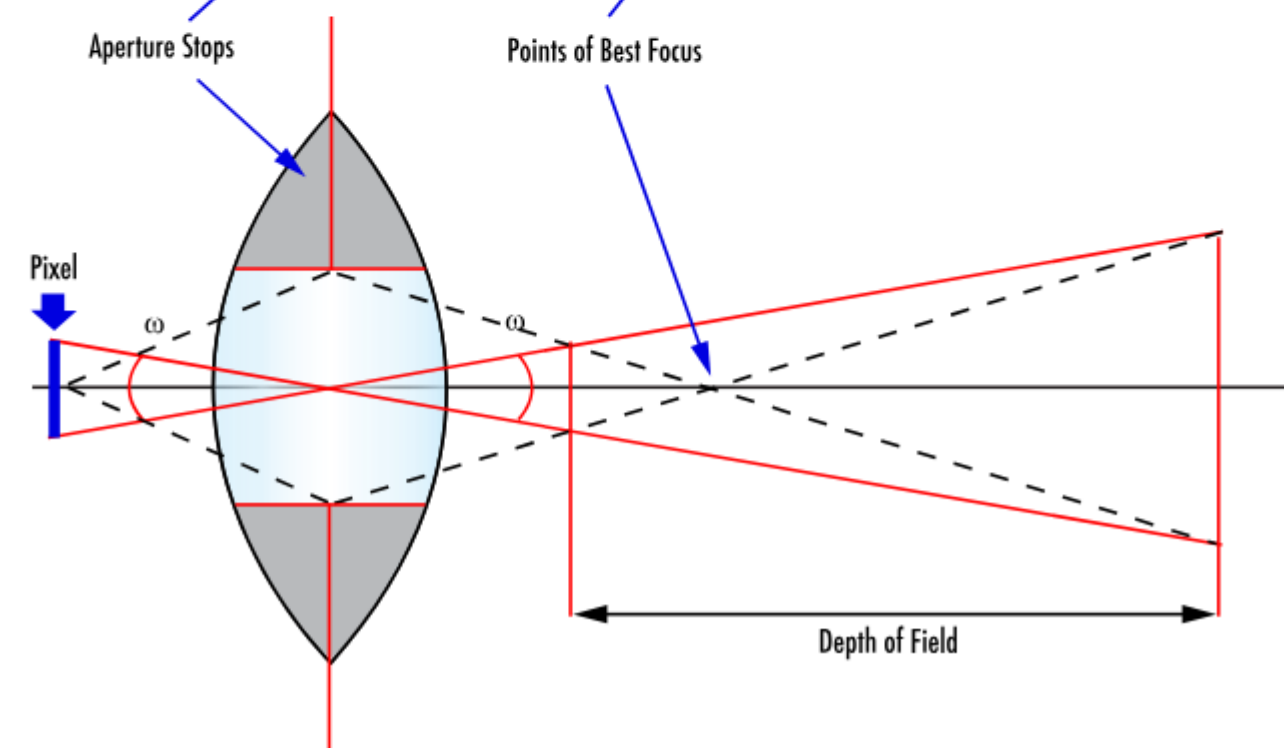
Componentes



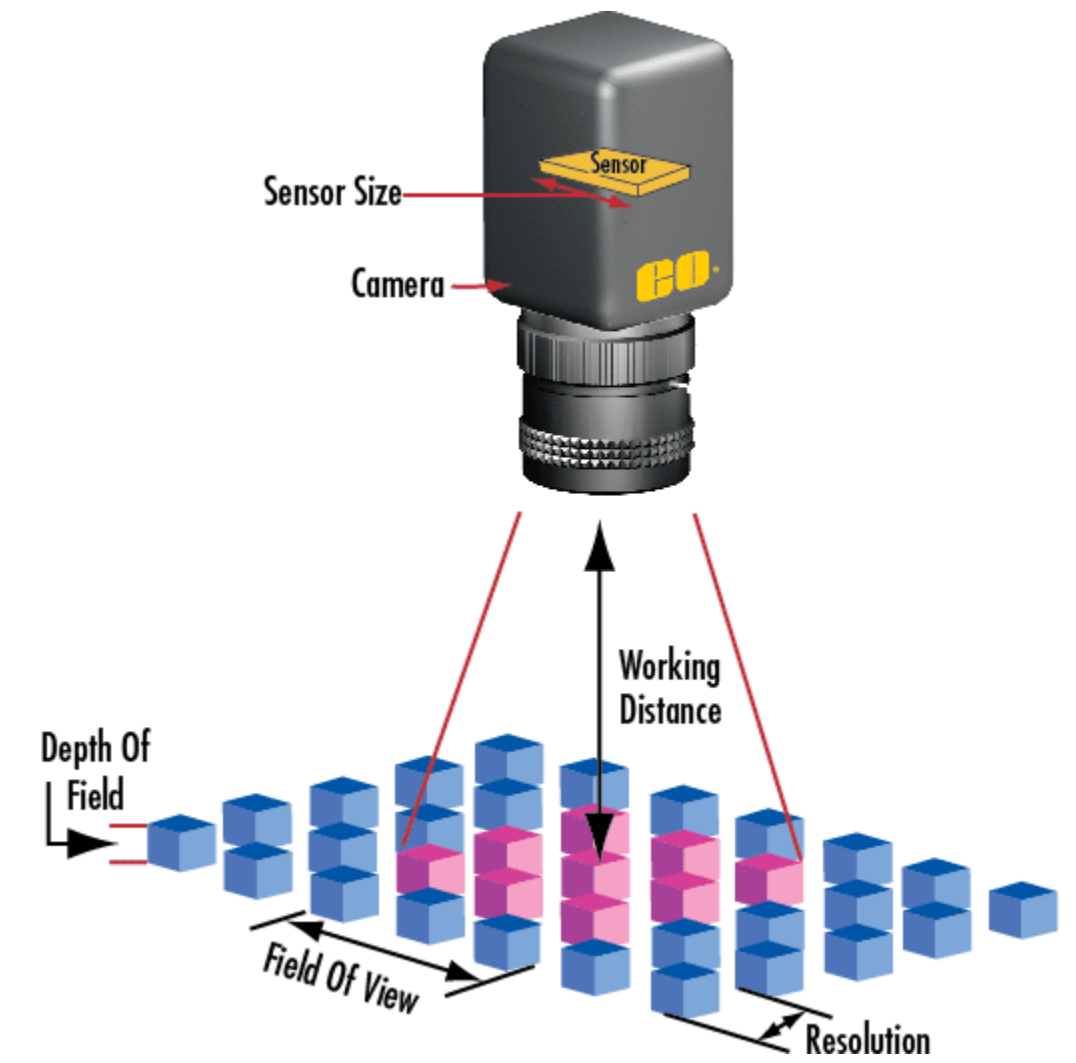
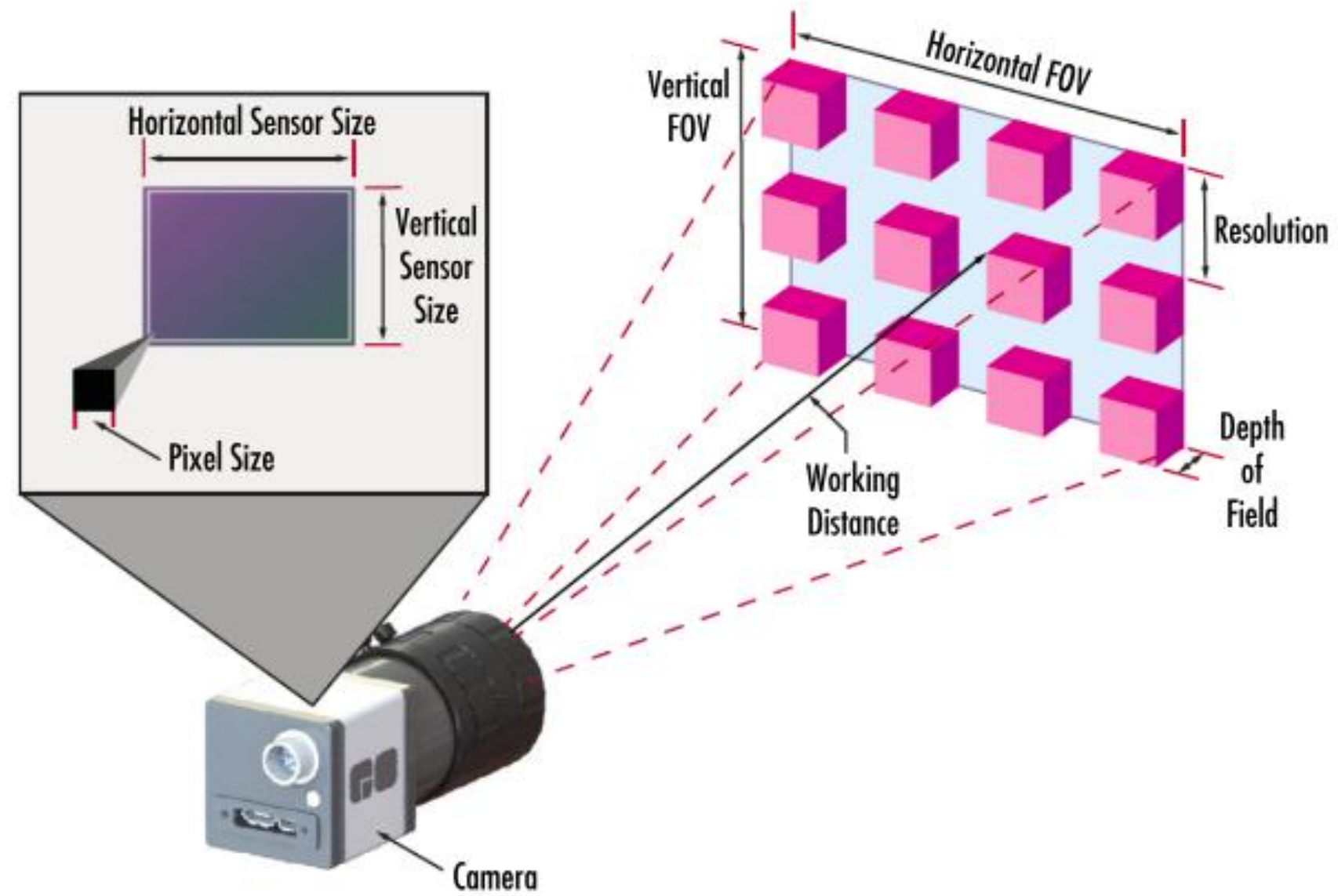
Low $f/\#$
Large Aperture



High $f/\#$
Small Aperture



Componentes

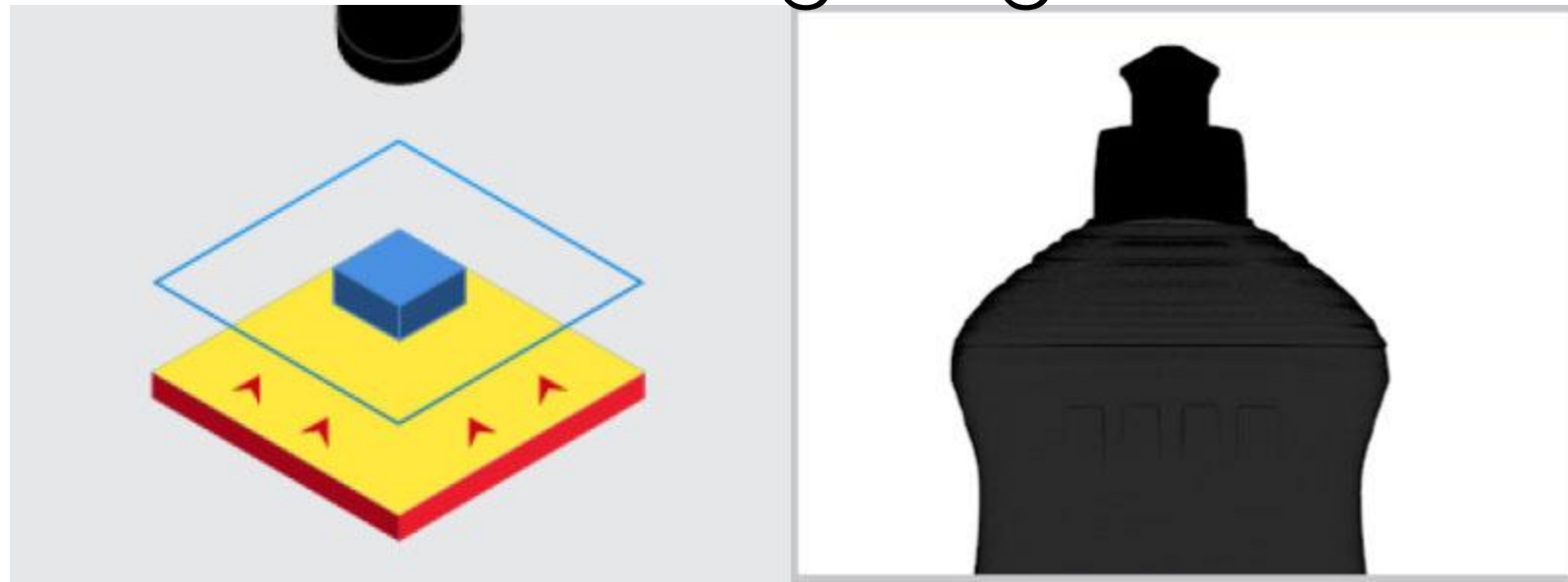


Componentes

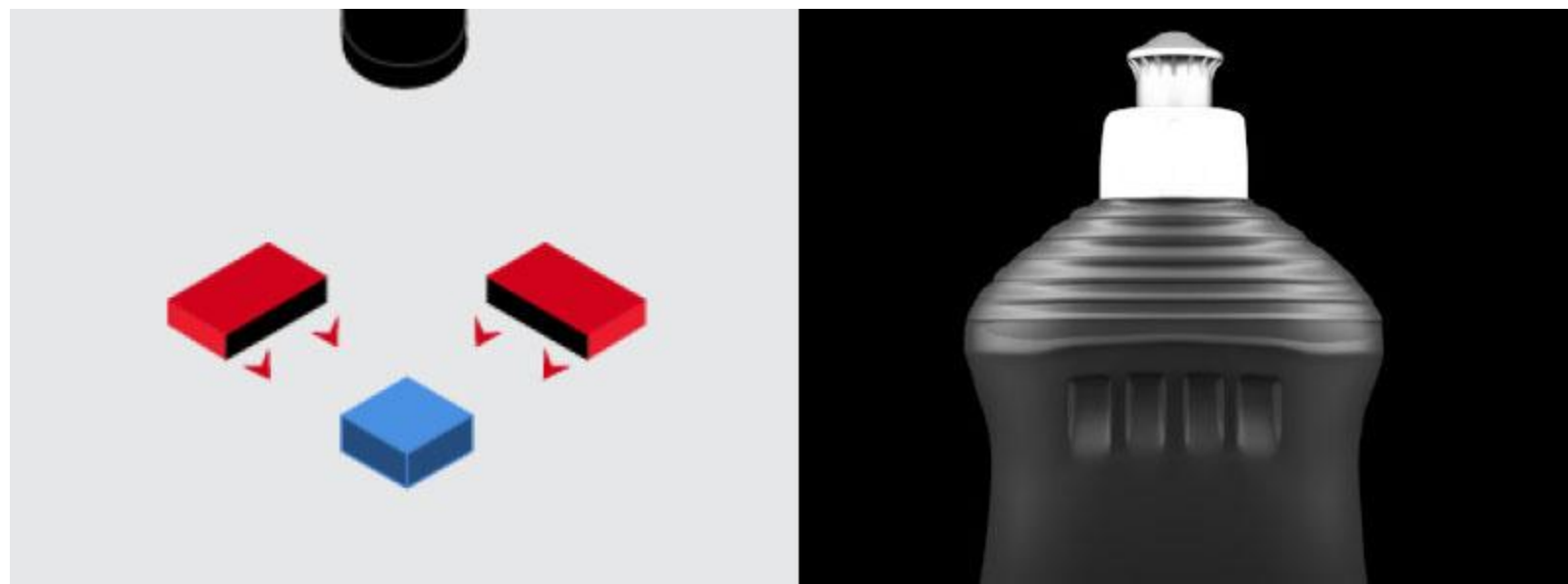
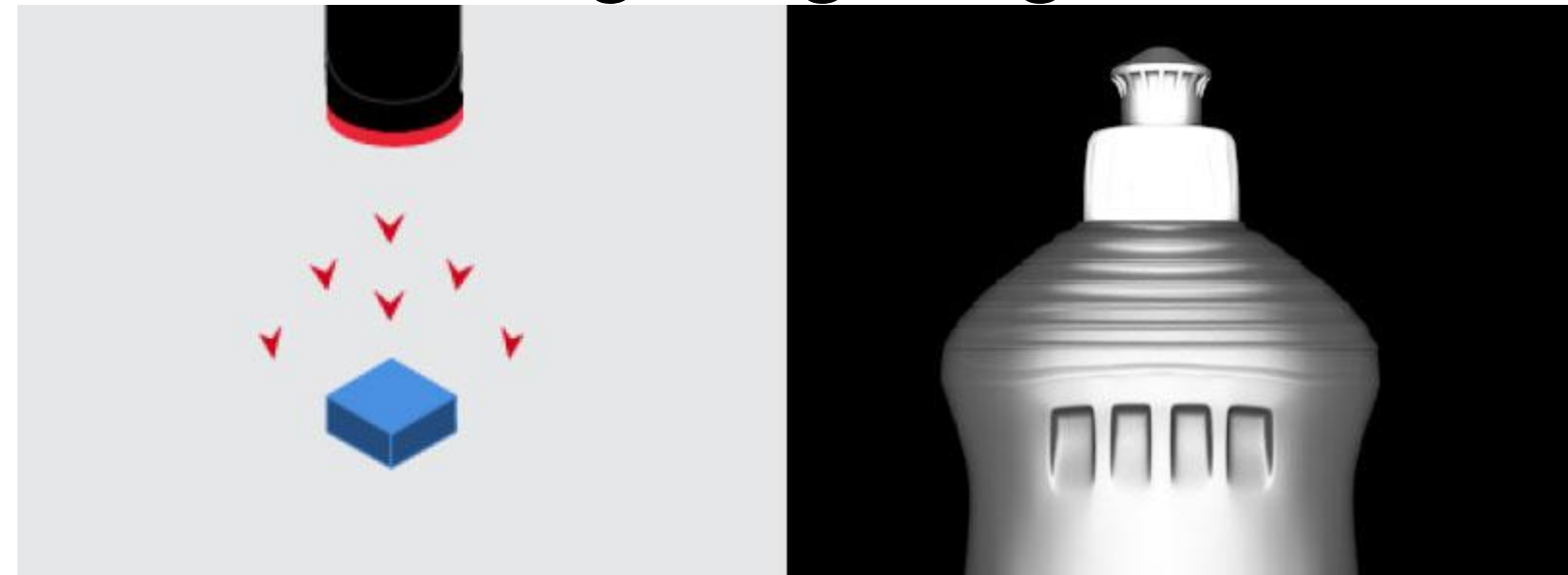


Sistema de iluminación

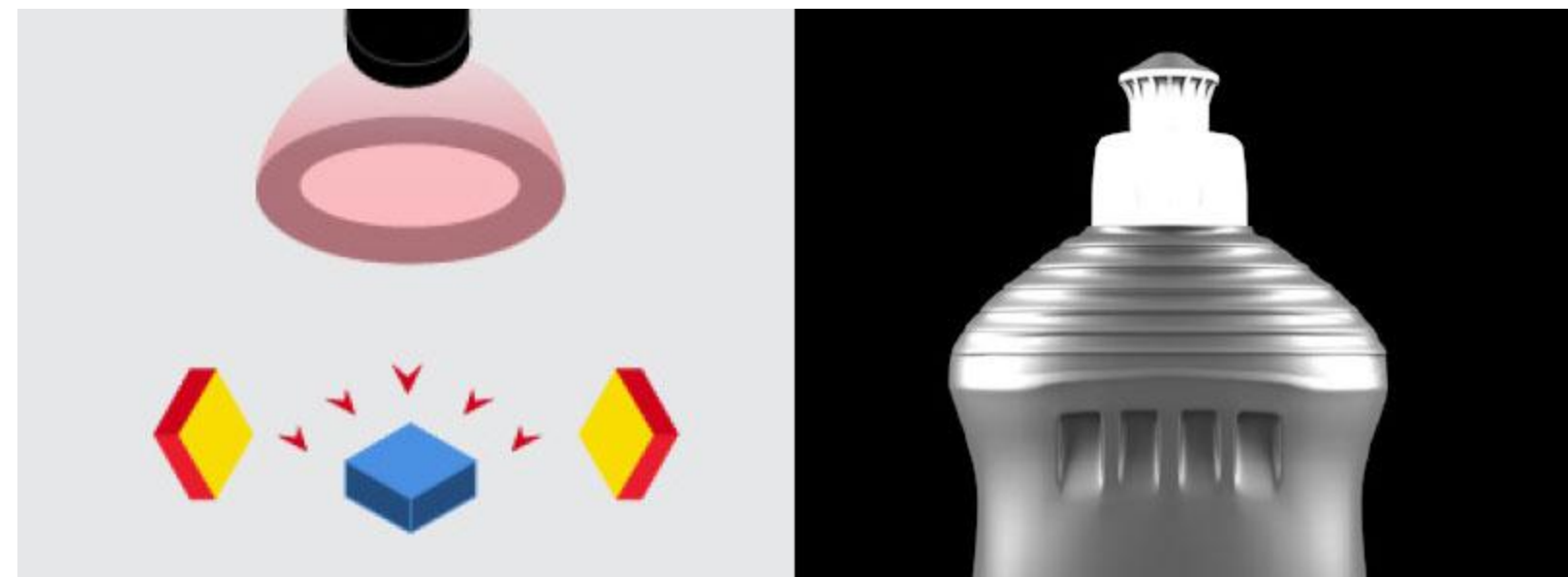
Backlighting



Bright lighting



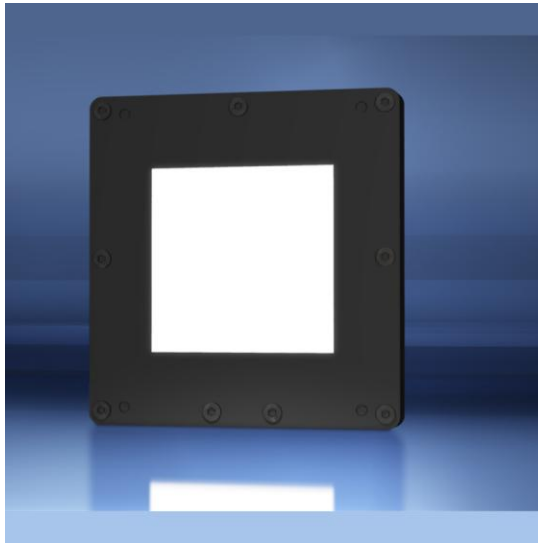
Dark field lighting



Diffuse lighting

Sistema de iluminación

Backlighting



Bright lighting



Dark field lighting



Diffuse lighting

Sistema de iluminación

